

FitBuddy – AI Fitness Plan Generator

Detailed Brainstorming & Ideation Phase

1. Introduction

The Brainstorming & Ideation phase is the first stage of the FitBuddy project. It identifies the problem, understands user needs, generates possible solutions, and defines the core idea of an AI-based fitness plan generator using Gemini Models.

2. Problem Identification

Many users find it difficult to create a suitable fitness routine on their own. A general workout plan may not match an individual's fitness goal, experience level, available time, or preferred exercise intensity. This creates a need for a system that can generate a personalized fitness plan from user-provided information.

3. Proposed Project Idea

FitBuddy is proposed as an AI-powered fitness plan generator. The system collects relevant user information and uses Gemini Models to generate a personalized fitness plan. The goal is to make fitness planning easier, accessible, and organized through a simple application interface.

4. Objectives Identified During Brainstorming

- Develop an easy-to-use fitness planning application.
- Generate personalized fitness plans from user inputs.
- Use Gemini Models to support AI-generated recommendations.
- Organize user information and generated plans clearly.
- Provide an administrator view for monitoring user-related information.
- Create a complete AI-based application suitable for demonstration.

5. Key Features Discussed

- User information collection such as age, weight, fitness goal, and workout intensity.
- AI-based generation of a fitness plan.
- Personalized workout recommendations based on supplied inputs.
- User plan history or plan viewing.
- Admin dashboard for viewing user information.
- Web interface using templates and static styling.

6. Gemini Models Integration – Initial Idea

Gemini Models are identified as the AI component of FitBuddy. The application can prepare a prompt from the user's fitness information and send it to the Gemini-based service. The generated response can then be presented as a structured fitness plan. Exact API configuration, prompt design, and implementation details can be finalized in later phases.

7. Proposed User Flow

Step	Activity
1	User enters relevant fitness information.
2	System validates and organizes the input.
3	Application prepares an AI prompt.
4	Gemini Models generate a fitness plan.
5	Generated plan is displayed to the user.
6	Plan information can be viewed or managed through the application.

8. Initial Technology Ideas

- Python as the primary programming language.
- Flask-style web application structure for the interface.
- HTML templates for web pages.
- CSS/static files for presentation and styling.
- Gemini Models for AI-assisted fitness-plan generation.

9. Expected Outcome

The project direction is defined: FitBuddy will focus on generating personalized fitness plans with the support of Gemini Models. The identified requirements and feature ideas provide the foundation for Requirement Analysis, Project Design, and Project Planning.

10. Summary

The brainstorming phase establishes FitBuddy as an AI-based fitness planning solution. The main concept is to combine user fitness information with Gemini Models to produce a personalized fitness plan through a web application. These ideas will be refined and implemented in the subsequent project phases.